Structural Fracture and Immunization Architecture

EU AI Act Articles 51–56

Final Version 3.1: Publication Edition

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AION System

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Survived three adversarial analysis passes and four independent Gauntlet verification reviews.

Document Status

Version: 3.1, Publication Edition

Status: Ready for Publication

Provenance: This document is the product of a three-pass adversarial analysis pipeline, followed by two independent Gauntlet reviews, a rework pass, and two independent Gauntlet Final Verification Passes. Both Final Verification Passes returned a Conditional verdict, together identifying two blocking issues and ten recommended disclosures. All twelve have been applied in this version. All quotes have been verified against primary sources. All external legislative claims have been independently confirmed.

Verification Record: 12 quotations verified against primary sources • 6 external legislative facts confirmed • 0 findings, resolutions, or thesis elements altered beyond what the two verification passes specifically identified.

A note on this page: the publication build corrects one claim from the working draft's process notes, which asserted a review history not reflected in the source document's own change log. See the note at the end of the Method section for what changed and why.

Executive Summary

This document fuses the structural findings from three independent adversarial passes against Articles 51–56 of the EU AI Act, as corrected following two independent Gauntlet reviews, a rework pass, and two independent Gauntlet Final Verification Passes that each returned a Conditional verdict. Analyzed through the PDE, TGA, MPTE, QAE, and SAR frameworks, the passes identified 18 surviving structural failure nodes; the severity distribution is 7 S4 and 11 S3 (see the Broader Finding Inventory). The Unified Thesis (that the Act commits an ontological category error by governing systemic cognitive infrastructure with product-safety mechanics) survived every Gauntlet pass unmodified and is retained as-is; this version adds a further external-review observation on the thesis's implications without altering the thesis itself.

This version applies two blocking fixes identified on the final verification pass (a self-contradicted cross-reference in Resolution 3, and a factually incorrect evidentiary-absence claim in the AI Office capacity disclosure) and ten recommended disclosures and clarifications addressing enforcement timing, peer-to-peer distribution, attestation auditing, trade-secret balancing, value-chain documentation gaps, capability-benchmark drift, staged-training aggregation, severity-methodology consistency, legislative-date precision, and the thesis's Treaty-basis implications. No finding, resolution, or thesis element was reopened beyond what the two verification passes specifically identified.

Method

This report was produced using the AION System adversarial pipeline, a six-stage process of structural analysis and external verification:

Foundry → Anvil → Crucible → Temper → Resolution → Gauntlet

Three full passes were run against Articles 51–56 of the EU AI Act. Each pass subjected the findings to independent adversarial review, negation testing, assumption archaeology, and severity calibration.

Killed findings: The process eliminated 6 weak findings across three passes. These are documented in the Killed List as proof of method.

External verification: The fused report underwent two independent Gauntlet reviews, a rework pass, and two independent Gauntlet Final Verification Passes. Both Final Verification Passes returned a Conditional verdict, together identifying two blocking issues and ten recommended disclosures; all twelve were applied in this patch (Version 3.1). No verification pass returned a Reject verdict.

Quote fidelity: All 12 direct quotations were independently re-extracted from primary sources and verified as exact matches.

External claims: Six external legislative facts were independently confirmed, including the AI Liability Directive withdrawal (6 October 2025), Directive (EU) 2024/2853's scope and application date, Regulation (EU) 2026/1744's enactment and provisions, and the GPAI Code of Practice finalization (10 July 2025).

Epistemic tags: Every claim in this report carries a tag indicating its basis:

[D] Directly observed or quoted from primary sources

[R] Reasoned inference from [D] evidence

[S] Strategic assessment or directional judgment

[?] Unverified or unknown

[M] Meta-commentary on the analysis itself

Severity ratings: Every finding carries a severity rating:

- **S4** Catastrophic: core obligation unenforceable
- **S3** Structural: weakness exploitable with consequence
- **S2** Operational: friction, recoverable
- **S1** Cosmetic: editorial or surface-level

Editorial note on this section. The working draft's build instructions specified a claim here of "six rounds of external hostile review by independent AI councils," with two Reject verdicts. That claim does not appear anywhere in the source document and contradicts its own change log, which records two Gauntlet reviews, a rework pass, and two Gauntlet Final Verification Passes: four review events, both Final Verification Passes CONDITIONAL, no Reject verdict recorded at any stage. This section states the reviewed history as the source document itself documents it.

1. The Unified Thesis

Core thesis unchanged: survived every Gauntlet pass; one addition on this pass.

The Final Upstream Failure Node: The Ontological Category Error

Regulating Systemic Cognitive Infrastructure using Product Safety Mechanics. [S]

The EU AI Act relies on the New Legislative Framework (CE marking, "placing on the market," "provider"). This framework assumes a discrete product with a fixed design, an identifiable manufacturer, and a point of sale. General-purpose AI models with systemic risk do not behave as products; they behave as environmental conditions or critical infrastructure. The hazard is not the transactional transfer of a model; the hazard is the achievement of systemic integration scale. Every downstream failure is a symptom of this upstream ontological mismatch. [S]

Retained caveat

[R] No resolution below fully escapes this frame: each attaches a new designated entity and product-safety-style obligation rather than replacing New Legislative Framework mechanics outright. The thesis is sharper than any of the four cures.

New corroborating evidence

[S] Regulation (EU) 2026/1744, enacted 27 July 2026, provides an unplanned real-world data point for this thesis: the EU's own legislature deferred and simplified precisely the New-Legislative-Framework/product-safety-style regime (Chapter III high-risk obligations) while leaving the systemic-risk GPAI regime (Chapter V, Articles 51–55) untouched in substance. The part of the Act built on discrete-product mechanics is the part currently under political pressure to loosen; the part built on a different logic is not. This does not prove the thesis, but it is evidence consistent with it that the report's authors could not have cited at the time of the original three passes. See the Political Feasibility section for the full discussion and its sourcing.

Further observation from external review

[S] [R] The Digital Omnibus's bifurcation (collapsing Chapter III product-safety mechanics while leaving Chapter V systemic-risk provisions untouched) suggests Chapter V may ultimately need governance closer to Digital Markets Act-style gatekeeper enforcement than to refined product-safety mechanisms. All four resolutions in this report operate within the New Legislative Framework paradigm the thesis itself critiques: each attaches a new designated entity to a product-safety-style obligation rather than replacing NLF mechanics outright (see the Retained caveat above). An infrastructure-governance model consistent with the thesis may require a different Treaty legal basis entirely: Article 173 TFEU or a sui generis instrument, rather than

Article 114 TFEU internal-market harmonisation, which can only regulate through economic operators. This is noted as a direction for future work, not a flaw in the current resolutions, and does not change the thesis or any resolution's mechanism in this version.

2. The Top Findings

Finding 1: The Open-Source Systemic Risk Enforcement Gap

S4: CATASTROPHIC

Enforcement unenforceable against non-compliant actors; not, as originally claimed, absent.

Articles: 53(2) • 54(1) and (6) • 3(3) • 101

Exact Text Quoted

> Art. 53(2): [D] "The obligations set out in paragraph 1, points (a) and (b), shall not apply to providers of AI models that are released under a free and open-source licence... This exception shall not apply to general-purpose AI models with systemic risks."
>
> Art. 54(1): [D] "Prior to placing a general-purpose AI model on the Union market, providers established in third countries shall, by written mandate, appoint an authorised representative which is established in the Union."
>
> Art. 54(6): [D] "The obligation set out in this Article shall not apply to providers of general-purpose AI models that are released under a free and open-source licence... unless the general-purpose AI models present systemic risks."
>
> Art. 3(3): [D] "'provider' means a natural or legal person... that develops... and places it on the market or puts the AI system into service under its own name or trademark..."

The finding as originally drafted claimed that the EU is left with no legal entity to bind to Chapter V obligations. That claim does not hold: Article 54(6) explicitly carves back into the representative requirement for exactly this case: open-source, systemic-risk models. Read together with 54(1), the Act does create a legal hook: a third-country provider of a systemic-risk model, open-source or not, must appoint an EU-based authorised representative.

The real gap is not textual absence: it is that Article 54(1) is a duty, not a self-enforcing gate. Nothing in the Act prevents a provider from simply not appointing a representative and continuing to release weights. Article 101 gives the Commission power to fine up to 3% of worldwide turnover or €15,000,000 [D] ("The Commission may impose on providers of general-purpose AI models fines not exceeding 3 % of their annual total worldwide turnover... or EUR 15 000 000, whichever is higher"), but a fine is only as good as the Commission's ability to collect it. Against a foreign, decentralized, or asset-free publisher with no EU presence and no representative to serve, the fine is a number on paper; the EU has no unilateral cross-border asset-seizure mechanism absent mutual legal assistance arrangements with the relevant third country. [R]

Severity justification. [R] The S4 rating is retained because the subset of actors this gap covers (foreign, judgment-proof, zero-EU-asset publishers of systemic-risk models) is precisely the population most likely to release the highest-capability models without any regard for EU regulatory authority. The gap is narrower than originally claimed, but it sits exactly where the catastrophic risk is highest: the actors an S4 rating exists to flag are, definitionally, the ones this enforcement gap cannot reach.

Note on enforcement timing. [D] The Commission's investigative and fining powers over GPAI providers under Article 101 (the enforcement lever this finding's corrected mechanism turns on) only became legally exercisable on 2 August 2026, one year after Chapter V's substantive obligations took effect on 2 August 2025. As of this patch pass the powers are live; the underlying argument (that a fine is only as good as the Commission's ability to collect it against a judgment-proof, non-EU actor) is unaffected by the exact activation date, but the report should have named it, and now does.

Severity-methodology consistency. [R] This S4 rating's retention despite a narrower population is acknowledged here as asymmetric with Finding 4's downgrade from S4 to S3 below: Finding 4 was stepped down because an ex officio backstop narrows its affected population. Finding 1's corrected gap is also narrower than originally claimed, yet keeps S4. The distinction that justifies the asymmetry: Finding 4 has a working backstop: ex officio designation eventually reaches the silent non-reporter. Finding 1 has no equivalent backstop for the judgment-proof foreign actor: the fine exists on paper, but no collection mechanism reaches a party with no EU assets and no representative to serve. The S4 rating therefore attaches to the absence of any enforcement backstop, not to the breadth of the affected population.

Why it matters. The gap is enforcement capacity against a non-compliant actor, not the absence of an obligation. This changes what a resolution needs to do: it doesn't need to invent a legal hook (one exists); it needs to make non-appointment costly or self-defeating before release, not punishable only after the fact against a party that may already be unreachable.

Finding 2: The Harmonisation and Evidentiary Gap in Ex-Post Harm Allocation

S3: STRUCTURAL

Retitled from "The Epistemic Gap of Ex-Post Harm Allocation."

Articles: 51–56 (absence of a Chapter-V-specific mechanism), read against Recital 9 and Directive (EU) 2024/2853

The finding as originally drafted claimed that Chapter V "contains zero mechanisms for ex-post harm allocation." That is factually incorrect as a zero-mechanisms claim. [D] Recital 9 of the Act states that "all rights and remedies provided for by such Union law to consumers, and other persons on whom AI systems may have a negative impact, including as regards the compensation of possible damages pursuant to Council Directive 85/374/EEC remain unaffected and fully applicable." Council Directive 85/374/EEC is the existing EU product liability regime, in force until 9 December 2026. [D] Separately, Directive (EU) 2024/2853 (the revised Product Liability Directive, already adopted) explicitly extends the definition of "product" to cover software and AI systems, applies to products placed on the market after 9 December 2026, and shifts the evidentiary burden toward defendants who cannot produce required technical documentation.

Scope caveat. [R] Recital 9's operative sentence sits in a paragraph that opens by discussing high-risk AI systems specifically, though its "without prejudice to existing Union law... to which this Regulation is complementary" framing is stated as a general principle applying across sectors. Whether courts will read Recital 9 as squarely covering Chapter V (GPAI-with-systemic-risk) harms with the same force as Chapter III (high-risk system) harms is not something the Act's text settles beyond doubt; this is an interpretive question, not a certainty. This caveat is directly relevant to Resolution 2's reframing below.

The mechanism is not absent: it exists via 85/374/EEC today and will strengthen via 2024/2853 from December 2026. What remains genuinely weak: (1) neither instrument is AI-specific: proving "defect" and causation for a general-purpose model's downstream harm inside a much longer AI value chain is harder than for a single-manufacturer physical product, and 2024/2853's evidentiary relief depends on the defendant possessing and being compellable to produce technical documentation, which loops back into Finding 1's enforcement problem for actors outside EU reach; (2) neither instrument creates a systemic catastrophic-harm compensation pool; (3) the "compliance shield" question (whether adherence to a Chapter V code of practice could be argued in litigation as evidence against a finding of "defect") is a real open legal question the Act does not resolve either way. [?]

Why it matters. This is a genuine, live gap: a harmonization-and-evidentiary-hurdles gap layered onto existing instruments, not the absence of any pathway to compensation the original finding claimed.

Finding 3: The Compound Regulatory Inertia Trap (Systemic Single Point of Failure)

S4: CATASTROPHIC

Articles: 52(4) • 56(9) • 90(2)

Exact Text Quoted

> Art. 52(4): [D] "The Commission may designate a general-purpose AI model as presenting systemic risks, ex officio or following a qualified alert from the scientific panel..."
>
> Art. 90(2): [D] "Upon such qualified alert, the Commission, through the AI Office and after having informed the Board, may exercise the powers laid down in this Section for the purpose of assessing the matter."
>
> Art. 56(9), full text: [D] "Codes of practice shall be ready at the latest by 2 May 2025... If, by 2 August 2025, a code of practice cannot be finalised, or if the AI Office deems it is not adequate following its assessment... the Commission may provide, by means of implementing acts, common rules..."

Currency note. Article 56(9), the specific provision this finding cites, is not among the paragraphs amended by Regulation (EU) 2026/1744 (the Digital Omnibus on AI, in force since 27 July 2026). The amending regulation's Article 1, point (21) amends only Article 56(6); see Resolution 3 below for the exact new text and what it means for that neighboring paragraph. Article 56(9)'s "may" language, quoted above, is confirmed current as of this pass [D], and this quotation of Article 56(9) is a quotation of current, unamended law: unlike Resolution 3's proposed future text below, it correctly retains "AI Office," since paragraph 9 (unlike paragraph 6) was not touched by the Digital Omnibus.

The GPAI Code of Practice was in fact finalized and published on 10 July 2025 [D], before the 2 August 2025 deadline. The specific scenario the original finding used to illustrate Article 56(9)'s trap (the Commission lets the deadline lapse, then declines to act) did not occur; the primary path succeeded. This half of the finding is updated accordingly, because the underlying "may" architecture around designation is untouched by this outcome.

Article 90(2) shows the discretion goes one step further back: even the Commission's decision to act on a qualified alert once received is itself "may," not "shall." This strengthens rather than weakens the systemic-single-point-of-failure argument: there are now two independent discretionary gates (receive-and-act, then designate), both convergent on Commission political will.

Severity justification. [R] S4 is retained, not merely carried over by default: both discretionary gates identified above (whether the Commission acts on a qualified alert at all, and whether it designates a model once it does act) sit on the single path by which a systemic-risk model not already self-reported gets brought under Chapter V's substantive obligations at all. A gate that a politically hesitant Commission can decline to open without breaching the letter of the law is not a friction point on an otherwise-working mechanism; it is the mechanism's only entry point failing to have a floor. That is what distinguishes this from an S3-level operational weakness: the failure mode is not "harder to enforce" but "never triggers."

Why it matters. The Compound Regulatory Inertia Trap survives, sharpened. Its Article 56(9) illustration needed replacing (the code-of-practice deadline was met), but its core claim (that a politically hesitant Commission can decline to exercise either the alert-response power or the designation power without breaching the letter of the law) is, if anything, understated in the original.

Finding 4: The Self-Assessment Silence Loophole

S3: STRUCTURAL

Downgraded from S4; retained at S3 on this pass.

Articles: 52(2) • 51(1)(a) and (b)

[D] Article 51(1)(b) already gives the Commission an independent designation path "based on a decision of the Commission, ex officio or following a qualified alert from the scientific panel"; fully separate from provider self-notification under 52(2). The scientific panel mechanism (Articles 68, 90) exists specifically to surface models the market itself under-reports. This is the same self-certification-plus-regulatory-backstop architecture used throughout EU law (e.g., GDPR breach notification, REACH), and it is qualitatively different from Finding 1 or Finding 2, where, pre-correction, no comparable backstop was in view. Rating this finding at the same severity as those two overstates it.

A provider that self-assesses below the Article 51(1)(a) benchmark can stay silent, forcing the Commission onto the resource-intensive ex officio path under 51(1)(b)/90. That remains true and remains a real design weakness (a self-reporting-first architecture will always be reactive at the margin), but it is a standard-grade structural weakness (S3), not a catastrophic core-obligation failure (S4), given the ex officio backstop the original finding itself acknowledged but did not weight.

3. The Resolution Architecture

Resolution 1: The Systemic Integration Controller

Addresses Finding 1

Proposed Text [S]: Insert new Article 54a.

> *Where a general-purpose AI model with systemic risk is made available for use within the Union, whether via the provision of access through application programming interfaces (APIs), the hosting of model weights for third-party download, or the provision of infrastructure for downstream fine-tuning, the obligations set out in Article 54(1) shall attach to whichever entity first meets the Systemic Integration Threshold defined below, in addition to (not in substitution for) the original provider's own obligations.*
>
> ***Systemic Integration Threshold:*** *an entity meets this threshold when, over any rolling 90-day period, it (a) routes or hosts access to the model for more than 500,000 monthly active end-users within the Union, OR (b) is a commercial entity (as opposed to a non-commercial, unincorporated community project) deriving revenue directly or indirectly from providing that access, OR (c) is itself designated by the AI Office following investigation as functioning as a de facto distribution chokepoint for the model within the Union, irrespective of raw user count, where evidence shows coordinated or affiliated operation across multiple nominally separate hosting or API entities designed to stay individually under threshold (a).*
>
> ***Safe harbor:*** *an entity that (i) hosts or routes access to the model, (ii) has fewer than 10,000 monthly active Union end-users, and (iii) is not commercially affiliated with the original provider or with any other entity meeting the Systemic Integration Threshold, is exempt from Article 54a obligations regardless of revenue status, provided it cooperates with AI Office information requests under Article 91.*
>
> ***Attestation requirement:*** *Entities claiming the safe harbor shall submit a signed attestation to the AI Office certifying their non-commercial status and lack of affiliation with the original provider or any Systemic Integration Controller. A false attestation shall be grounds for designation as a Systemic Integration Controller and shall constitute an infringement under Article 101.*
>
> ***Attestation audit trigger:*** *The AI Office shall conduct random or risk-based audits of safe-harbor attestations, prioritizing entities whose integration patterns or traffic volumes suggest threshold proximity. Failure to respond to an attestation audit request within 30 days shall void the safe-harbor exemption and trigger designation review under threshold prong (c).*

The attestation's infringement basis was corrected during drafting: an earlier draft cited Article 99, the Member-State-administered general penalties provision (fines up to €35M/7%, €15M/3%, or €7.5M/1%, for prohibited-practice and general provider/deployer/importer/distributor infringements) [D]. Article 54a sits in Chapter V, the GPAI systemic-risk regime, which is centrally enforced by the Commission under Article 101, the same provision Finding 1's own corrected mechanism already relies on for the €15M/3% GPAI fine tier. The citation is corrected to Article 101 above for internal consistency.

"Systemic integration scale" is replaced with a defined threshold (user count, commercial-activity status, and an anti-fragmentation clause for coordinated slicing) rather than an undefined phrase. [S] The specific numbers (500,000 users, 10,000-user safe harbor) are illustrative policy proposals, not derived from the Act; they would need empirical calibration before real legislative drafting. [?] The revised text attaches obligations to weight-hosting-for-download as well as API access, specifically to close the "release raw weights, let others self-host" escape route. The safe harbor exempts small hobbyist mirrors and API wrappers below 10,000 monthly users from commercial affiliates.

Disclosed limitation. [R] If every entity in the chain is genuinely non-EU, non-commercial, and individually under-threshold, this resolution still cannot reach them. A subsequent Gauntlet pass classified this specific disclosure as misdirected-as-an-attack rather than a genuine wound: Article 3(65)'s own definition requires "systemic risk" to mean impact "having a significant impact on the Union market due to their reach," so an entity with genuinely zero EU nexus is close to definitionally incapable of posing the risk Chapter V exists to catch in the first place, and Article 2(1)(a)/(c)'s extraterritorial hooks already capture most models that would meet that bar. This limitation is retained here for completeness, but its practical scope is narrower than the original phrasing suggested.

Additional disclosed limitation. [R] Purely peer-to-peer distribution (a magnet link seeded by independent peers, an IPFS swarm, or comparable decentralized protocols) has no single "hosting" entity and no identifiable chokepoint that meets any of the three Systemic Integration Threshold prongs above. A systemic-risk model distributed via a fully decentralized swarm of thousands of independent, uncoordinated peers may achieve systemic EU-wide reach without any single entity crossing the threshold, since prong (c)'s coordination requirement presumes a designer of the fragmentation, and spontaneous, uncoordinated mirroring has none. This is a jurisdictional and technical limit of the regional statutory approach (the same category of limit as the zero-EU-nexus point above), not a drafting defect within this resolution's own terms.

Legal Basis. [R] Article 114 TFEU.

Classification. Architecturally resolved for EU-nexus, server-based actors; thresholds require empirical calibration before enactment, and peer-to-peer distribution remains outside this resolution's reach (see disclosed limitations above).

Resolution 2: Closing the Evidentiary Gap

Addresses Finding 2, retitled from "Piercing the Compliance Shield"

The original resolution's legal basis, a parallel AI Liability Directive, is dead. [D] The proposal (2022/0303/COD) was formally withdrawn by the European Commission; the withdrawal notice was published in the Official Journal (C/2025/5423) on 6 October 2025. That vehicle no longer exists and cannot be the basis for a live resolution.

The two sub-paragraphs below were originally presented as equally load-bearing. They are not. Sub-paragraph (2) is the operative fix: it amends Chapter V directly and creates a binding documentation-disclosure obligation. Sub-paragraph (1) is a supporting interpretive clarification. EU recitals are non-binding interpretive aids: they inform how a court reads an operative article, but a recital amendment does not, on its own, create a new binding legal obligation or override what the operative articles actually say. The proposed text below reflects this: (1)'s language states an interpretive direction rather than asserting a legal outcome, and the resolution states explicitly that (2) carries the binding weight.

Proposed Text [S]

*(1) **AI Act Recital Amendment**, interpretive clarification, not independently binding: Add to Recital 9 a clarification that Directive (EU) 2024/2853's technical-documentation-based evidentiary presumptions shall be interpreted as applying to providers and authorised representatives of general-purpose AI models with systemic risk, and that adherence to a Chapter V code of practice does not, by itself, constitute a defence to a finding of defect under Directive (EU) 2024/2853. This paragraph guides judicial interpretation of the operative provisions below; it does not itself create a binding obligation independent of them.*

*(2) **Chapter V amendment**, the operative fix: Providers of general-purpose AI models with systemic risk shall maintain, and make available to claimants' counsel upon a substantiated request certified by a national court, the technical documentation required under Article 53(1)(a) and Annex XI, subject to trade-secret protections under Article 78, closing the gap where a claimant cannot invoke Directive 2024/2853's documentation-based defect presumption because the AI-specific documentation regime (Article 53) and the general product-liability documentation-disclosure regime (2024/2853) are not currently linked in text. The binding force of this resolution rests on this paragraph, not on paragraph (1).*

(3) For genuinely cross-border catastrophic harms exceeding [X] EUR[a] in aggregate claims from a single systemic-risk model incident, Member State courts hearing the lead claim may refer questions of causation and damage aggregation to a dedicated AI Office technical panel for a non-binding expert opinion, addressing the black-box causation problem 2024/2853 only partially solves for frontier general-purpose models.

[a] Placeholder: requires empirical calibration before enactment.

Damage/liability boundary. This resolution deliberately does not propose new strict, uncapped liability. Given Directive 2024/2853 already provides a no-fault-adjacent, defect-based regime with claimant-favorable evidentiary rules, adding a second, undefined "cascading failure / systemic blast radius" strict liability layer on top risks uninsurable tail-risk exposure and provider flight from the EU market. The resolution instead links and clarifies existing instruments rather than creating new uncapped exposure.

Acknowledged limitation. [R] Article 78(1)(a) explicitly protects intellectual property rights and confidential business information, including source code. A national court hearing a documentation-disclosure request under paragraph (2) above will need to balance the claimant's evidentiary need against the provider's trade-secret shield. This resolution creates the disclosure mandate but does not resolve the judicial balancing test between disclosure and trade-secret protection; that remains an open interpretive question for national courts, not something a Chapter V amendment can settle in advance.

Acknowledged limitation. [R] Where a downstream EU entity assumes provider status under Article 25 through substantial modification or rebranding, that entity may not possess the original base-model Annex XI documentation held by the foreign upstream provider; it may hold only the weights, not the pre-training data curation and evaluation logs Annex XI requires. The disclosure mandate in paragraph (2) above binds the legal provider of record, but the upstream foreign entity that actually holds the underlying documentation may remain outside the court's jurisdiction and outside this resolution's reach. This gap is a manifestation of Finding 1's enforcement problem (an EU-reachable defendant without the evidence, and an evidence-holding defendant outside EU reach), not an independent defect of this resolution.

Legal Basis. [R] Recital amendment (interpretive) + Chapter V cross-reference amendment to Article 53 (operative), the latter procedurally lighter than a new Directive; Article 114 TFEU. **Classification.** Resolved via linkage to Directive (EU) 2024/2853; the operative mechanism is paragraph (2), not paragraph (1); subject to the trade-secret balancing and value-chain documentation limitations disclosed above.

Resolution 3: Linked Trigger and Mandatory Fallback

Addresses Finding 3

This resolution was rebased against Regulation (EU) 2026/1744. The exact operative text of the amendment to Article 56, obtained directly from the Official Journal (OJ L, 2026/1744, 24.7.2026, Article 1, point 21), reads: [D]

> "in Article 56, paragraph 6 is replaced by the following: '6. The Commission and the Board shall regularly monitor and evaluate the achievement of the objectives of the codes of practice by the participants and their contribution to the proper application of this Regulation. The Commission, taking utmost account of the opinion of the Board, shall assess whether the codes of practice cover the obligations provided for in Articles 53 and 55, and shall regularly monitor and evaluate the achievement of their objectives. The Commission shall publish its assessment of the adequacy of the codes of practice."'

Two things follow directly from this text: first, only Article 56(6) was amended: Article 56(9), the paragraph this resolution targets, does not appear anywhere in Regulation (EU) 2026/1744's Article 1 amendment list (which runs to 32 numbered points touching dozens of other articles by name) and is therefore confirmed unchanged. [D] Second, the amendment removed the Commission's implementing-act power to give a code of practice general validity (the sentence granting that power is deleted from paragraph 6 entirely) and shifted paragraph 6's assessment/monitoring language from "the AI Office and the Board" to "the Commission and the Board."

Political-feasibility implication. This resolution's core ask (turn the Article 56(9) fallback from "may" to "shall") now runs directly against the grain of a decision the real legislature just made one paragraph over. Recital 41 of Regulation (EU) 2026/1744 states [D] that the Article 56(6) implementing-act power was removed because "it is not strictly necessary for these codes to be approved by an implementing act," given codes of practice have "limited legal effect." That is a deregulatory move in the opposite direction from this resolution's proposal to make the neighboring implementing-act fallback mandatory. This does not make Resolution 3 wrong: Article 56(9)'s fallback serves a different purpose, a backstop for when the primary path fails, versus 56(6)'s general approval mechanism. But it is disclosed here because a resolution proposing new mandatory Commission duties should not omit that the same legislature, in the same amending act, to the same article, moved the other way on a structurally similar power. [R]

Proposed Text [S] (date-handling clause tightened)

> ***Amend Article 56(9):*** *If, by 2 August 2025 (historical date retained for text continuity; a future equivalent amendment would use a forward-looking deadline), a code of practice cannot be finalised, or if the Commission deems it is not adequate following its assessment under paragraph 6, and such deeming determination shall itself be made within 60 days of the relevant assessment being due, the Commission* ***SHALL****, not may, provide, by means of implementing acts, common rules for the implementation of the obligations provided for*

in Articles 53 and 55, within 6 months of the deeming determination.

Note on cross-reference currency: *paragraph 6's assessment function, referenced above, is now performed by "the Commission, taking utmost account of the opinion of the Board" rather than "the AI Office and the Board," per Article 1, point (21) of Regulation (EU) 2026/1744. This amendment should track that actor going forward.*

Amend Article 52(4)/90(2): *Upon a qualified alert under Article 90(1), the Commission shall, within 30 days, either (a) initiate designation proceedings under Article 52(4), or (b) issue a reasoned public decision under Article 296 TFEU explaining why it is declining to do so. Where proceedings are initiated under (a), the Commission shall conclude the designation determination within 90 days of initiation, absent a reasoned public extension of up to 60 further days.*

The proposed text above now reads "if the Commission deems," matching the cross-reference note that follows it, correcting a discrepancy in an earlier draft, where the classification line claimed this cross-reference had been updated when the underlying text still read "the AI Office deems."

The original text mandated only "initiate... within 30 days," with no obligation to ever conclude. The revision adds a 90-day conclusion deadline (with one bounded, disclosed extension), closing the meta-loophole where discretion could relocate from "whether to start" to "how long to take." It also adds a deadline to the "deeming" step itself (60 days), closing the gap where the six-month clock could never start.

Legal Basis. [R] Article 56(9) / 52(4) / Article 296 TFEU (duty to provide reasoned decisions). **Classification.** Resolved: both delay points closed; the cross-reference to Article 56(6)'s current actor is corrected to "the Commission" in the proposed text and confirmed current as of Regulation (EU) 2026/1744.

Resolution 4: Reaching the Silent Non-Reporter

Addresses Finding 4, retitled and redesigned

The original resolution triggered an independent audit only when a provider sought an exemption from Article 51(1)(a) designation. Finding 4 is specifically about the provider who never enters that process at all. The resolution is redesigned around a trigger the silent provider cannot avoid by staying silent.

Proposed Text [S]

*(1) **Mandatory Compute-Threshold Registration:** Amend Article 52 to add: any entity that trains a general-purpose AI model using cumulative training compute exceeding* 10^{24} *FLOPs shall register that fact with the AI Office within 30 days of training completion, regardless of the entity's own assessment of whether Article 51(1)(a) is met. Registration triggers no immediate compliance obligation beyond the fact of registration itself; the Commission retains sole authority to make the systemic-risk designation under 51(1) and 52(4).*

Aggregation clarification: *For the purposes of this registration requirement, "cumulative training compute" shall aggregate across all staged, incremental, continual, or fine-tuning training runs applied to the same base model or its direct descendants within a 12-month rolling window, regardless of whether any single run individually exceeds the threshold, closing the gap where staged sub-threshold training tranches could otherwise avoid ever triggering a single "completion" event.*

(2) The 10^{24} *FLOP registration trigger sits one order of magnitude below the Act's own* 10^{25} *FLOP presumption threshold under Article 51(2), close enough to catch models approaching the line, without sweeping in the much larger population of mid-sized commercial models a* 10^{23} *threshold (two orders below) would have captured. This is a registration-only obligation, not a full compliance or audit trigger, which further limits the burden relative to the original* 10^{23} *proposal.*

Delegated-act linkage: *The* 10^{24} *FLOP registration trigger shall be subject to the same delegated-act update mechanism under Article 97 as the Article 51(2) presumption threshold, ensuring both thresholds are recalibrated in tandem as algorithmic efficiency and hardware capability evolve. Article 51(3) already empowers the Commission to amend the Article 51(2) marker "in light of evolving technological developments, such as algorithmic improvements or increased hardware efficiency"; this registration trigger should not be pinned to a static number when the threshold immediately above it is explicitly designed not to be.*

*(3) **Independent Audit for Exemption Claims** (retained, unchanged in mechanism): Any provider of a general-purpose AI model exceeding the Article 51(2) presumption threshold must submit a capability assessment verified by an AI-Office-accredited independent third party to claim exemption from Article 51(1)(a) designation.*

*(4) **Whistleblower/Third-Party Alert Channel:** Amend Article 90 to add a paragraph enabling any natural or legal person, including downstream deployers and employees of a provider, to submit a substantiated alert to the AI Office regarding a general-purpose AI model they reasonably believe meets Article 51(1) criteria and has not been registered under (1) above, with confidentiality and retaliation protections modeled on Directive (EU) 2019/1937 (the EU Whistleblower Directive).*

Substantiation standard: *For the purposes of this paragraph, a "substantiated alert" from a non-technical alerter requires only a good-faith, factually specific account of the model's deployment context and observed behavior, not a technical capability assessment; the AI Office shall evaluate the technical merit of the alert* *independently.*

(5) ***Capability Benchmark Matrix:*** *The* 10^{24} *FLOP registration trigger under (1) shall be paired with a dynamic capability benchmark matrix (e.g., MMLU, MMMU, or successor benchmarks), evaluated by the Scientific Panel under Article 68. The FLOP threshold addresses training-scale visibility; the benchmark matrix addresses actual capability, since algorithmic efficiency gains (mixture-of-experts architectures, distillation, synthetic data curricula) mean a model trained below the FLOP threshold can match or exceed the capabilities of an earlier model trained well above it. Where a model trained below the FLOP threshold demonstrates benchmark performance indicating potential Article 51(1)(a) capabilities, the Commission may still designate it ex officio under Article 52(4), independent of registration status under (1).*

Institutional-capacity caveat. Public staffing data exists: the AI Office reports 125+ staff, with the Commission confirming 140+ total and 40+ in the AI Safety unit as of June 2026 [D] (Commission parliamentary reply to MEP Piotr Müller, reported by Agence Europe, 9 June 2026; both figures independently reconfirmed against the AI Office's own site and the Agence Europe bulletin on this pass), but whether this is sufficient for the audit capacity implied by (3) and the benchmark evaluation implied by (5) at scale remains unresolved. [?] A third-party assessment (Pour Demain, May 2026, via Lawfare) found current levels "inadequate" relative to enforcement demands and recommended scaling to at least 160 by 2030. This resolution does not itself solve that capacity gap.

Legal Basis. [R] Article 52(2) amendment; Article 90 amendment; Article 97 delegated-act linkage; Article 68 Scientific Panel linkage; Directive (EU) 2019/1937 as a model for whistleblower protections.

Classification. Resolved for detection, including a capability-benchmark backstop against FLOP-threshold gaming; institutional capacity for the audit and benchmark-evaluation components remains an open implementation risk, disclosed with sourced current data above.

4. Political Feasibility

This section previously described an open, monitorable political situation. It no longer is one: the situation has resolved, and this section states the resolved fact rather than tracking it as pending.

[D] Regulation (EU) 2026/1744, the "Digital Omnibus on AI," was signed on 8 July 2026, published in the Official Journal on 24 July 2026, and entered into force on 27 July 2026. The underlying political agreement was reached on 7 May 2026; the European Parliament adopted its position on 16 June 2026, and the Council approved the regulation on 29 June 2026, completing the ordinary legislative procedure [D]. It amends Regulation (EU) 2024/1689 across 32 numbered points, most consequentially deferring Chapter III high-risk AI system obligations from 2 August 2026 to 2 December 2027 (Annex III standalone systems) and 2 August 2028 (Annex I embedded systems). The political question this report's prior version treated as open (will the Commission's simplification-and-delay agenda touch Chapter V?) is now answerable directly from the enacted text rather than as a forecast.

The answer, sourced directly from the amending regulation: [D] Chapter V (Articles 51–55, the systemic-risk GPAI provisions this report is built around) is untouched in substance. Article 1 of Regulation (EU) 2026/1744 amends dozens of named articles (1, 2, 3, 4, 5, 6, 10, 11, 17, 25, 27, 28, 29, 30, 40, 42, 43, 50, 56, 57, 58, 60, 63, 64, 69, 70, 72, 75, and inserts new Articles 4a, 60a, 75a) and Articles 51 through 55 do not appear among them. The one point of contact with Chapter V is Article 56(6) (see Resolution 3 above for the exact amended text), which governs Codes of Practice monitoring and removes a now-superfluous implementing-act power; it does not touch Articles 51–55's systemic-risk thresholds, provider obligations, or the "may" architecture Finding 3 attacks.

Disclosed honestly, not treated as vindication. [S] Every resolution in this report proposes to add obligations (a new controller category, technical-documentation linkage, faster deadlines, registration requirements, whistleblower channels) at a moment when the enacted political direction (confirmed, not forecast) is toward delay and simplification for the Chapter III regime this report doesn't address, and toward removing rather than adding Commission implementing-act power in at least one Chapter V paragraph (Article 56(6)) directly adjacent to a provision (Article 56(9)) this report proposes to make more demanding. That specific tension is disclosed in Resolution 3 above rather than here, since it is resolution-specific rather than report-wide.

External corroboration for the Unified Thesis. [S] The enactment of Regulation (EU) 2026/1744 provides external corroboration of the Unified Thesis stated above: the EU chose to delay and simplify the product-safety-style high-risk regime (Chapter III) while leaving the systemic-risk GPAI regime (Chapter V) substantively untouched. The part of the Act built on New Legislative Framework mechanics is under political pressure; the part built on a different logic is not. This is not proof of the thesis: a single legislative episode is not a controlled experiment, and the deferral's stated rationale (delayed standards, heavier-than-expected compliance burden, per Recital 40) is about implementation readiness, not about the product-safety-versus-infrastructure distinction this report's thesis turns on, but it is a real, citable, and unplanned data point in the thesis's favor, and this report claims it as one rather than leaving it unmentioned.

This report's resolutions should be read as a technical readiness document, not as proposals with current legislative momentum. Nothing in Regulation (EU) 2026/1744 creates an opening for any of Resolutions 1–4; nothing in it forecloses one either. The honest position, now that the relevant political question is settled rather than pending, is that these resolutions remain exactly as ready, or unready, for legislative pursuit as their own internal design makes them, independent of the Digital Omnibus.

5. The Broader Finding Inventory

Total findings discovered across three passes: 18. The 14 findings outside the top four remain undisclosed in this fused artifact; [?] this version does not resurrect or newly disclose them.

Category	Count
Total findings	18
S4 (Catastrophic)	7
S3 (Structural)	11
Killed findings	6
Undisclosed findings	14

6. The Killed List

Survived every Gauntlet pass, including this one, presented as proof of method.

1. **Delegation of Core Obligations to Codes of Practice** (Pass 1): misread of the safe-harbor mechanism.
2. **"Without Undue Delay" Reporting Ambiguity** (Pass 1): standard EU drafting artifact, not a structural fracture.
3. **Good Faith Safe Harbor for Downstream Deployers** (Pass 2): creates moral hazard and an intentional-ignorance loophole.
4. **Temporal Rot in Reassessment Lock-In** (Pass 2/3): misread of administrative queue versus enforcement powers.
5. **Provisional Acceptance for Article 53(4) upon 90-day delay** (Pass 3): massive regulatory capture vector; enables a legalized shadow market.

A sixth entry is carried in the inventory as a cumulative artifact of the negation engine across all three passes, rather than as a named, independently killed finding.

7. Remaining Open Items

1. **Illustrative numeric thresholds.** [S][?] The illustrative numeric thresholds in Resolution 1 (500,000 / 10,000 users) and the Resolution 2(3) damage-claim aggregation figure ([X] EUR[1]) remain placeholders, not resolved to specific numbers. Neither verification pass had the empirical basis (EU user-base data by model, litigation-cost data) to specify a defensible number, and inventing one to look complete would itself be a fabrication.

2. **AI Office institutional capacity.** [D][?] Resolution 4(3) and (5)'s institutional-capacity question is now disclosed with sourced data, not fully resolved. Public staffing figures exist (AI Office: 125+ staff per its own site; Commission-confirmed 140+ total, 40+ AI Safety, June 2026); a third-party assessment found current levels inadequate relative to enforcement demand and recommended 160 by 2030. Whether that is sufficient for economy-wide independent audit and benchmark-evaluation of exemption/registration claims at scale is a judgment call this report cannot make from documentary sources alone: the data exists now, the adequacy answer doesn't.

3. **Extraterritorial-enforcement limit.** [R] The extraterritorial-enforcement limit on Resolution 1 (an entity with zero EU nexus at all, including via genuinely uncoordinated peer-to-peer distribution) remains disclosed as a jurisdictional and technical limit. No drafting choice inside a regional regulation can compel a party with genuinely no EU nexus, or reach a distribution swarm with no coordinating entity; this is a jurisdictional and technical fact, not a defect this document can draft around.

4. **Killed List and undisclosed findings.** The Killed List and the 14 undisclosed findings in the Broader Inventory remain unreopened, per explicit instruction across all passes including this one.

5. **Trade-secret balancing and upstream documentation.** [?] Trade-secret balancing under Resolution 2(2) and the upstream-documentation gap under value-chain rebranding (Resolution 2) are disclosed as open interpretive/jurisdictional questions, not resolved. Neither is answerable from the source documents alone; both require either judicial interpretation (trade secrets) or a jurisdictional fact pattern this document cannot draft around (upstream documentation holder outside EU reach).

Everything else the two verification passes flagged (the Resolution 3 cross-reference, the AI Office capacity claim, and all ten recommended disclosures) has been applied and is no longer open.

[1] Placeholder: requires empirical calibration before enactment.

8. Verification Record

Quote fidelity. All 12 direct quotations in this report were independently re-extracted from primary sources and verified as exact matches: 9 from the text of Regulation (EU) 2024/1689 (the EU AI Act), and 3 from Regulation (EU) 2026/1744's Official Journal text (the Article 56(6) replacement text, and two separate fragments of Recital 41). This corrects a count carried over from the working draft, which stated 11 and 1; recounted directly against the quotations actually reproduced in this document on this pass. No quotation required an unverified tag.

External claims confirmed (6).

1. The AI Liability Directive's formal withdrawal (proposal 2022/0303/COD; withdrawal notice OJ C/2025/5423, 6 October 2025).
2. Directive (EU) 2024/2853's scope (extending "product" to cover software and AI systems) and its 9 December 2026 application date.
3. Regulation (EU) 2026/1744's enactment timeline: signed 8 July 2026, published in the Official Journal 24 July 2026, in force 27 July 2026.
4. Regulation (EU) 2026/1744's substantive provisions, including the exact amended text of Article 56(6) and confirmation that Articles 51–55 are untouched.
5. The GPAI Code of Practice's finalization and publication on 10 July 2025.
6. AI Office and Commission AI Safety unit staffing figures (125+ / 140+ / 40+, June 2026) and the Pour Demain third-party capacity assessment (May 2026), both independently reconfirmed against their original sources on this pass, not merely carried forward from the prior draft.

No fabricated figures. The illustrative numeric thresholds in Resolution 1 (500,000 / 10,000 monthly users) and the Resolution 2(3) damage-claim aggregation figure are retained as explicit placeholders rather than presented as derived or empirically calibrated values, consistent with the Remaining Open Items disclosure above.

Primary sources. Regulation (EU) 2024/1689 (EU AI Act), Articles 2, 3, 25, 51–56, 90–92, 101, and Recital 9; Regulation (EU) 2026/1744 (Digital Omnibus on AI), fetched from the Official Journal (EUR-Lex); the withdrawal notice for proposal 2022/0303/COD (OJ C/2025/5423); Directive (EU) 2024/2853; Directive (EU) 2019/1937; public AI Office staffing disclosures, including a Commission parliamentary reply to MEP Piotr Müller reported by Agence Europe (9 June 2026) and the AI Office's own website; the Pour Demain AI Office capacity assessment via Lawfare (May 2026).

Epistemic tag legend

[D] Directly observed / quoted

[R] Reasoned inference

[S] Strategic / directional

[?] Unverified / unknown

[M] Meta-commentary

Severity rating legend

- S4 Catastrophic
- S3 Structural
- S2 Operational
- S1 Cosmetic

Appendix: Revision History

Summarized from the working document's internal change log and patch log.

v1.0 Original fused report. Three independent adversarial passes against Articles 51–56 produced 18 surviving findings (7 S4, 11 S3) and a killed list of 6.

v2.0 / v3.0 Post-Gauntlet correction and rework passes (the source document's own change log tracks these two stages as one combined table). Legislative currency updated against the enacted Digital Omnibus; Resolution 3 rebased against the Official Journal's exact amendment text; Resolution 2's interpretive-versus-operative split clarified after the parallel AI Liability Directive was confirmed withdrawn; Resolution 1's attestation and audit mechanisms added; Resolution 4's compute threshold linked to the Article 97 delegated-act mechanism; Finding 1's S4 severity given an explicit justification; a corroboration paragraph added to the Unified Thesis.

v3.1 Post-patch pass. Two blocking fixes applied: a self-contradicted cross-reference in Resolution 3, and a factually incorrect evidentiary-absence claim in the AI Office capacity disclosure, both caught on the second Gauntlet Final Verification Pass. Ten recommended disclosures added: enforcement-timing precision for Finding 1, a peer-to-peer distribution limitation for Resolution 1, an attestation audit trigger for Resolution 1, a trade-secret balancing caveat and an upstream-documentation caveat for Resolution 2, a capability benchmark matrix and a staged-training aggregation clarification for Resolution 4, a severity-methodology consistency note reconciling Finding 1 and Finding 4, a legislative-date precision correction in Political Feasibility, and a Treaty-basis observation added to the Unified Thesis. All twelve fixes applied; none declined.

AION System

Foundry → Anvil → Crucible → Temper → Resolution → Gauntlet

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"I take no sides. I read what the document actually says."